

Supplementary material to:

Optimizing autonomous multimodal last-mile delivery systems with time windows: analyzing trade-offs between drones, robots, and trucks

A. Valid inequalities for the MILP formulations

To improve the performance of the MILP formulation, we introduce a set of valid inequalities in Equations (1) - (5).

$$d_{n+1,k}^T \geq \sum_{i \in V_{\mathcal{L}}} \sum_{\substack{j \in V_{\mathcal{R}}, \\ j \neq i}} t_{ij}^T x_{ijk}^T + \sum_{i \in V} (d_{ik}^T - a_{ik}^T) \quad k \in \mathcal{K} \quad (1)$$

$$d_{n+1,k}^T \geq \sum_{p \in \mathcal{P}} \sum_{i \in V_{\mathcal{L}}} \sum_{\substack{j \in V_{\mathcal{N}}, \\ j \neq i}} \sum_{\substack{h \in V, \\ h \neq j}} (l^D + t_{ij}^D + \bar{l}^D + t_{jh}^D) x_{ijh}^D \quad k \in \mathcal{K} \quad (2)$$

$$\sum_{k \in \mathcal{K}} \sum_{p \in \mathcal{P}} \sum_{i \in V_{\mathcal{L}}} \sum_{\substack{j \in V_{\mathcal{N}}, \\ j \neq i}} \sum_{\substack{h \in V, \\ h \neq j}} x_{ijh}^D + \sum_{k \in \mathcal{K}} \sum_{p \in \mathcal{P}} \sum_{r \in \mathcal{R}} \sum_{i \in V_{\mathcal{L}}} \sum_{\substack{j \in V_{\mathcal{N}}, \\ j \neq i}} x_{ijr}^R \leq n \quad (3)$$

$$\sum_{k \in \mathcal{K}} \sum_{j \in V_{\mathcal{H}}} x_{0jk}^T \geq 1 \quad (4)$$

$$\sum_{k \in \mathcal{K}} \sum_{i \in V_{\mathcal{L}}} x_{ijk}^T \leq 1 \quad j \in V_{\mathcal{H}} \quad (5)$$

Equations (1) - (2) define a lower bound on the final departure time from the depot $\{n+1\}$, i.e., the makespan. Although trucks do not leave the depot $\{n+1\}$, it is necessary to compute the truck departure times $d_{n+1,k}^T$ to encompass drone sorties that finish at $\{n+1\}$ after the arrival of trucks. Equation (3) provides a lower bound for the maximum number of delivery operations that drones and robots can perform. Equation (4) states that at least one truck should leave the depot. Equations (5) limit the number of trucks that can visit each node. Note that Equations (4) and (5) are not necessary to model the VRPD-RS-TW, but they improve the flow balance of the truck arcs in the linear relaxation.

Tables 1 - 3 present the results of the MILP formulations without (MILP) and with valid inequalities (MILP_{Val}). Table 1 shows results including drivers' wages, while Table 2 excludes them. Additionally, Table 3 reports the results for the makespan objective. Each table reports the best objective value found (Z), the best bound (BB), the optimality gap (%), and computational time in seconds. The best average results are highlighted in bold.

Overall, the inclusion of valid inequalities (1) - (5) does not lead to significant performance improvements in most cases. The most notable difference is observed in the operational cost objective when drivers' wages are considered (Table 1), where MILP_{Val} shows better average objective values, tighter bounds, and smaller optimality gaps. For the operational cost objective when drivers' wages are excluded (Table 2), both MILP and

MILP_{Val} consistently find optimal solutions across all instances. Nevertheless, the addition of valid inequalities helps achieve optimal solutions in shorter computational times. Then, for the makespan objective function, MILP and MILP_{Val} present similar results for instances of 5 and 15 customers, whereas MILP_{Val} shows lower incumbents and gaps for the instances of 10 customers. Therefore, despite the modest improvement in overall MILP_{Val} performance, we choose to include valid inequalities in the MILP formulations due to the observed enhancements in both solution quality and computational efficiency, particularly for instances with 10 and 15 customers in Table 1. These results are presented and discussed in **Section 5.2.1**.

Table 1: Comparison of MILP and MILP_{Val} when including the drivers' wages in the objective function.

Instance		MILP				MILP _{Val}			
#	<i>n</i>	Z	BB	Gap [%]	Time [s]	Z	BB	Gap [%]	Time [s]
1	5	132.23	132.23	0.00	33.89	132.23	132.23	0.00	29.47
2		57.35	57.35	0.00	11.14	57.35	57.35	0.00	7.97
3		151.39	151.39	0.00	35.89	151.39	151.39	0.00	34.17
1	10	65.19	6.69	89.73	7203.77	64.32	26.73	58.44	7204.58
2		135.54	6.47	95.23	7230.31	134.01	20.64	84.60	7204.19
3		120.12	5.04	95.80	7204.28	96.90	16.14	83.34	7225.36
1	15	163.28	6.50	96.02	7221.27	122.47	6.54	94.66	7224.05
2		170.94	8.63	94.95	7208.28	169.44	21.26	87.46	7205.47
3		146.34	4.87	96.67	7214.95	132.75	8.01	93.96	7216.13
Avg		126.93	42.13	63.16	4818.20	117.87	48.92	55.83	4816.82

Table 2: Comparison of MILP and MILP_{Val} when excluding the drivers' wages in the objective function.

Instance		MILP				MILP _{Val}			
#	<i>n</i>	Z	BB	Gap [%]	Time [s]	Z	BB	Gap [%]	Time [s]
1	5	4.30	4.30	0.00	4.69	4.30	4.30	0.00	0.48
2		6.00	6.00	0.00	0.92	6.00	6.00	0.00	1.19
3		3.13	3.13	0.00	0.89	3.13	3.13	0.00	1.08
1	10	5.85	5.85	0.00	729.70	5.85	5.85	0.00	847.72
2		6.64	6.64	0.00	568.44	6.64	6.64	0.00	484.34
3		3.99	3.99	0.00	138.03	3.99	3.99	0.00	43.52
1	15	4.73	4.73	0.00	1179.73	4.73	4.73	0.00	1262.66
2		8.50	8.50	0.00	823.66	8.50	8.50	0.00	334.75
3		4.43	4.43	0.00	713.05	4.43	4.43	0.00	309.19
Avg		5.28	5.28	0.00	462.12	5.28	5.28	0.00	364.99

Table 3: Comparison of MILP and MILP_{Val} for the makespan objective function.

Instance		MILP				MILP _{Val}			
#	<i>n</i>	Z	BB	Gap [%]	Time [s]	Z	BB	Gap [%]	Time [s]
1	5	4.24	4.24	0.00	37.20	4.24	4.24	0.00	28.06
2		1.47	1.47	0.00	9.22	1.47	1.47	0.00	1.02
3		4.94	4.94	0.00	41.59	4.94	4.94	0.00	28.48
1	10	1.47	0.00	100.00	7203.58	1.44	0.54	62.67	7203.69
2		4.16	0.00	100.00	7208.77	4.16	0.39	90.72	7201.72
3		2.43	0.00	100.00	7204.00	2.33	0.36	84.71	7209.13
1	15	5.20	0.00	100.00	7214.30	3.18	0.00	100.00	7218.94
2		5.22	0.00	100.00	7220.34	5.22	0.00	100.00	7215.31
3		3.92	0.00	100.00	7225.02	3.30	0.00	100.00	7241.02
Avg		3.67	1.18	66.67	4818.23	3.37	1.33	59.79	4816.37

B. Feasibility checks of the 3P-GMS-ILS metaheuristic

In the 3P-GMS-ILS, candidate solutions are rejected if one of the following equations is violated. For simplicity, we skip the sub-indexes corresponding to trucks, drones, and robots. Drone and robot feasibility checks are defined in Equations (6) and (7).

$$a_h^D - d_i^D \leq L_{ijh}^D \quad i \in V_{\mathcal{L}}, j \in V_{\mathcal{N}}, h \in V_{\mathcal{R}} \quad (6)$$

$$l^R + t_{ij}^R + \bar{l}^R + t_{ji}^R \leq T \quad p \in \mathcal{P}, v \in \mathcal{R}, i \in V_S, j \in V_{\mathcal{N}} \quad (7)$$

Time window feasibility checks are covered by Equations (8) - (9) for trucks, Equations (10) for drone multi-legs, Equations (11) for drone round-trips, and Equations (12) for robot deliveries.

$$a_i^T \leq s_i^T \leq d_i^T \quad i \in V_{\mathcal{N}} \quad (8)$$

$$t_i^{start} \leq s_i^T \leq t_i^{stop} \quad i \in V_{\mathcal{N}} \quad (9)$$

$$t_i^{start} \leq a_i^D \leq t_i^{stop} \quad i \in V_{\mathcal{N}} \quad (10)$$

$$t_j^{start} \leq r_{ip}^D + l^D + t_{ij}^D \leq t_j^{stop} \quad p \in \mathcal{P}, i \in V_{\mathcal{L}}, j \in V_{\mathcal{N}} \quad (11)$$

$$t_j^{start} \leq r_{ip}^R + l^R + t_{ij}^R \leq t_j^{stop} \quad p \in \mathcal{P}, i \in V_S, j \in V_{\mathcal{N}} \quad (12)$$

Being S a subset of customers in a truck route, the trucks' capacity is checked by Equations (13).

$$\sum_{i \in S} q_i \leq \mathcal{Q} \quad S \subseteq V_{\mathcal{N}} \quad (13)$$

C. Parameter Calibration of the 3P-GMS-ILS Metaheuristic

To calibrate the 3P-GMS-ILS metaheuristic, we applied the non-parametric Friedman test (Oda et al., 2015) to identify significant differences, for the input parameters ranging between $R \in \{5, 10, 15, 20, 25\}$ and $\beta \in \{1.0, 1.5, 2.0, 2.5, 3.0, 3.5\}$, over 15 representative instances. We performed 10 independent runs per instance, normalizing objective values to account for varying instance sizes. Then we applied the post hoc Nemenyi test (Liu and Chen, 2012) for pairwise comparison. Table 4 shows the mean ($\bar{x}$), standard deviation (σ), standard error (SE), and confidence intervals (CI) of the different parameter configurations.

The results show that the best configuration of parameters is found for values of $R = 20$ and $\beta = 3.0$, which yielded a mean normalized objective of 1.0594 (SD = 0.1199) and a 95% confidence interval of [0.9930, 1.1258]. The Friedman test returned a small p-value ($p \approx 1.0561374 \cdot 10^{-23}$), confirming highly significant differences among the parameter configurations. Therefore, the configuration: $R = 20$ and $\beta = 3.0$ is adopted for all the paper's experiments.

Table 4: Statistical analysis of 3P-GMS-ILS metaheuristic for different parameter configurations of R and β .

Parameters (R, β)	$\bar{x}$	σ	SE	CI Lower	CI Upper
(5, 1.0)	1.2012	0.1916	0.0495	1.0951	1.3073
(5, 1.5)	1.1981	0.1909	0.0493	1.0924	1.3038
(5, 2.0)	1.1539	0.1716	0.0443	1.0589	1.2490
(5, 2.5)	1.1352	0.1539	0.0397	1.0499	1.2204
(5, 3.0)	1.1082	0.1660	0.0429	1.0162	1.2001
(5, 3.5)	1.1373	0.1752	0.0452	1.0403	1.2343
(10, 1.0)	1.1316	0.1561	0.0403	1.0452	1.2181
(10, 1.5)	1.1237	0.1396	0.0360	1.0463	1.2010
(10, 2.0)	1.1094	0.1464	0.0378	1.0283	1.1904
(10, 2.5)	1.1091	0.1571	0.0406	1.0221	1.1961
(10, 3.0)	1.1050	0.1678	0.0433	1.0120	1.1979
(10, 3.5)	1.0921	0.1552	0.0401	1.0062	1.1781
(15, 1.0)	1.1265	0.1430	0.0369	1.0473	1.2057
(15, 1.5)	1.0918	0.1294	0.0334	1.0202	1.1635
(15, 2.0)	1.0819	0.1486	0.0384	0.9996	1.1642
(15, 2.5)	1.0737	0.1416	0.0366	0.9953	1.1521
(15, 3.0)	1.0677	0.1228	0.0317	0.9996	1.1357
(15, 3.5)	1.0718	0.1276	0.0329	1.0012	1.1425
(20, 1.0)	1.1055	0.1437	0.0371	1.0260	1.1851
(20, 1.5)	1.0955	0.1374	0.0355	1.0195	1.1716
(20, 2.0)	1.0774	0.1528	0.0395	0.9928	1.1620
(20, 2.5)	1.0747	0.1214	0.0313	1.0075	1.1419
(20, 3.0)	1.0599	0.1187	0.0307	0.9942	1.1256
(20, 3.5)	1.0668	0.1410	0.0364	0.9888	1.1449
(25, 1.0)	1.1075	0.1444	0.0373	1.0275	1.1875
(25, 1.5)	1.0821	0.1310	0.0338	1.0096	1.1546
(25, 2.0)	1.0741	0.1525	0.0394	0.9896	1.1585
(25, 2.5)	1.0606	0.1027	0.0265	1.0038	1.1175
(25, 3.0)	1.0594	0.1199	0.0310	0.9930	1.1258
(25, 3.5)	1.0668	0.1135	0.0293	1.0039	1.1296

D. Simulated annealing metaheuristic framework

Algorithm 1 describes the Simulated Annealing (SA) metaheuristic framework implemented in the experiments of Section 5.2.2. The input parameters are the transportation time matrices of trucks, drones, and robots (t^T, t^D , and t^R), maximum computational time (δ_T), cooling ratio (α), initial temperature ($temp_i$), and final temperature ($temp_f$). The output is the best solution found during exploration S^* . In addition, t^c represents the annealing temperature, and S' is a temporal variable that stores the neighbors of the current solution S_i . Furthermore, the counter i determines the current iteration, $f(\cdot)$ provides the objective function value of the solutions, t^i is the initial computational time, and t^f is the final computational time. Regarding the parameters that determine the performance of the SA, in pre-computational experiments we tested the performance of SA under all the configurations for the following parameters: $temp_i \in \{800, 900, 1000\}$, $temp_f \in \{1, 5, 10\}$, and $\alpha \in \{0.997, 0.998, 0.999\}$. Then, we performed 10 replications for each instance from the Amsterdam set. The best performance of SA was achieved under the following configuration: $temp_i = 1000$, $temp_f = 1$, and $\alpha = 0.999$. Therefore, the results presented in Section 5.2.2 correspond to this configuration of parameters. In addition, to provide a fair comparison with the 3P-GMS-ILS, the maximum number of neighbors explored within every neighborhood is equal to $|V_N|^2$.

The initialization phase starts by establishing the default values for the initial time (t_i), the annealing temperature (t^c), the stopping condition (*stopping_condition*), the counter of iterations (i), and creating an initial VRPD-RS-TW solution (lines 1 - 3). The initial solution is created by function **Phase_I**($\cdot$) (see Section 4.1.1) and stored in S_i and S^* (line 3). Then, in the iterative phase, the algorithm performs the exploration of neighborhood set $\mathcal{N}(\cdot)$ until either the maximum computational time or the final temperature is reached (lines 4 - 22). Within every iteration, a for-loop explores every neighborhood $\mathcal{N}_l$ (line 6). Every neighborhood $\mathcal{N}_l$ is ordered in an increasing size. Then, the neighbors of S_i are stored in S' (line 7), and S_i is updated based on the SA acceptance criterion (lines 8 - 14). After that, S^* is updated in case the solution in S_i outperforms the incumbent S^* (lines 15 - 16). Once the algorithm leaves the for-loop, the annealing temperature is adjusted (line 17) and the final computational time is measured (line 18). With this, the termination criterion is evaluated for the maximum computational time and the final temperature (lines 19 and 20). Then the algorithm updates S_{i+1} and the counter i . Finally, the algorithm returns the best solution found during exploration S^* .

Algorithm 1: SA

Data: ($t^T, t^D, t^R, \delta_T, \alpha, temp_i, temp_f$)

1 Initialization Phase:

2 $t_i \leftarrow time(), t^c \leftarrow temp_i, stopping_condition \leftarrow False, i \leftarrow 0;$

3 $S^*, S_i \leftarrow \mathbf{Phase_I}(t^T, t^D, t^R);$

4 Iterative Phase:

5 while not stopping_condition do

6 for $l \in |\mathcal{N}|$ do

7 $S' \leftarrow \mathcal{N}_l(S_i);$

8 if $f(S') < f(S_i)$ then

9 $S_i \leftarrow S';$

10 else

11 $\Delta \leftarrow f(S_i) - f(S');$

12 $r \leftarrow random(0, 1);$

13 if $r < exp(\frac{\Delta}{t^c})$ then

14 $S_i \leftarrow S';$

15 if $f(S_i) < f(S^*)$ then

16 $S^* \leftarrow S_i;$

17 $t^c = t^c \cdot \alpha;$

18 $t^f \leftarrow time();$

19 if $t^f - t^i \geq \delta_T$ or $t^c \leq temp_f$ then

20 $stopping_condition \leftarrow True;$

21 $S_{i+1} \leftarrow S_i;$

22 $i \leftarrow i + 1;$

Result: S^*

E. Statistical tests for metaheuristic comparisons

To evaluate the performance of the 3P-ILS, 3P-GILS, and 3P-GMS-ILS metaheuristics, we benchmark these metaheuristics in Section 5.2.2 across the makespan and operational-costs objective functions. To assess the statistical significance of the metaheuristics' performance, we first normalized the objective values, as the instance sizes vary considerably. Specifically, for each instance, we divided the objective value of each solution by the best performance observed across all algorithms and runs for that instance. This normalization enables fair comparisons across instances with different scales. Table 5 presents the summary of normalized performance metrics across both the makespan and operational-costs objective functions, where the columns show the mean,

standard deviation (std), standard error of the mean (SE), and the bounds of the 95% confidence interval (CI).

The averaged normalized objective values were used to perform the Friedman test, which assesses whether the differences in performance among the metaheuristics are statistically significant. For the operational costs objective function, the Friedman test yielded a p-value of 0.0004, below 0.05, indicating significant differences among 3P-ILS, 3P-GILS, and 3P-GMS-ILS. In contrast, for the makespan objective, the Friedman test reported a p-value of 0.1353, exceeding 0.05, suggesting no significant differences in performance. To further investigate the differences, the Nemenyi post-hoc test was applied for pairwise comparisons. In the operational costs, the results show that 3P-GMS-ILS significantly outperforms both 3P-ILS and 3P-GILS ($p < 0.05$), while no significant differences were found between 3P-ILS and 3P-GILS. These findings confirm that 3P-GMS-ILS consistently achieves the lowest normalized objective values and demonstrates superior performance in operational costs, with the lowest mean, median, standard deviation, and the narrowest confidence intervals. On the other hand, the similar performance across all metaheuristics for the makespan objective suggests that each method is equally effective for minimizing this objective function.

Table 5: Significance analysis for the 3P-ILS, 3P-GILS, and 3P-GMS-ILS metaheuristics.

	3P-ILS	3P-GILS	3P-GMS-ILS
3P-ILS	1.000000	0.984803	0.012796
3P-GILS	0.984803	1.000000	0.020920
3P-GMS-ILS	0.012796	0.020920	1.000000

(a) Pairwise Comparison for Operational Costs

	3P-ILS	3P-GILS	3P-GMS-ILS
3P-ILS	1.000000	0.871308	1.000000
3P-GILS	0.871308	1.000000	0.871308
3P-GMS-ILS	1.000000	0.871308	1.000000

(b) Pairwise Comparison for Makespan

	mean	std	SE	CI lower	CI upper
3P-ILS	1.1326	0.1893	0.0446	1.0384	1.2267
3P-GILS	1.1230	0.1760	0.0415	1.0354	1.2105
3P-GMS-ILS	1.0329	0.0688	0.0162	0.9987	1.0671

(c) Statistical Analysis for Operational Costs

	mean	std	SE	CI lower	CI upper
3P-ILS	1.0000	0.0000	0.0000	1.0000	1.0000
3P-GILS	1.0004	0.0014	0.0003	1.9997	1.0010
3P-GMS-ILS	1.0000	0.0000	0.0000	1.0000	1.0000

(d) Statistical Analysis for Makespan

To assess the performance of the General Variable Neighbourhood Search (GVNS), Simulated Annealing (SA), and 3P-GMS-ILS metaheuristics, we benchmark these methods in **Section 5.2.3** across the makespan and operational cost objective functions. The experimental setup mirrors that of **Section 5.2.2**. Table 6 follows the same format as Table 5, presenting results for the comparison between GVNS, SA, and 3P-GMS-ILS.

The averaged normalized objective values were analyzed using the Friedman test to assess whether the performance differences among the metaheuristics were statistically significant. The Friedman test yielded p-values of 0.000006 and 0.000009 for the operational cost and makespan objectives, respectively, both below 0.05, indicating significant differences between GVNS, SA, and 3P-GMS-ILS. To identify where these differences occurred, the Nemenyi post-hoc test was applied for pairwise comparisons. The results show that 3P-GMS-ILS significantly outperforms both GVNS and SA ($p < 0.05$ for each objective), while no significant difference was observed between GVNS and SA. Additionally, 3P-GMS-ILS exhibited the narrowest confidence intervals, indicating high certainty in its estimated performance. Overall, 3P-GMS-ILS consistently achieved the lowest normalized objective values and outperformed the other metaheuristics in terms of mean, median, standard deviation, and confidence interval width, demonstrating its robustness across the tested instances.

Table 6: Significance analysis for the GVNS, SA, and 3P-GMS-ILS metaheuristics.

	GVNS	SA	3P-GMS-ILS
GVNS	1.000000	0.871308	0.000063
SA	0.871308	1.000000	0.000519
3P-GMS-ILS	0.000063	0.000519	1.000000

(a) Pairwise Comparison for Operational Costs

	GVNS	SA	3P-GMS-ILS
GVNS	1.000000	0.524419	0.000092
SA	0.524419	1.000000	0.005805
3P-GMS-ILS	0.000092	0.005805	1.000000

(b) Pairwise Comparison for Makespan

	mean	std	SE	CI lower	CI upper
GVNS	1.5026	0.4891	0.1153	1.2593	1.7458
SA	1.5713	0.4275	0.1008	1.3587	1.7838
3P-GMS-ILS	1.0329	0.0688	0.0162	1.9987	1.0671

(c) Statistical Analysis for Operational Costs

	mean	std	SE	CI lower	CI upper
GVNS	1.4883	0.7439	0.1753	1.1183	1.8582
SA	1.3649	0.4854	0.1144	1.1235	1.6062
3P-GMS-ILS	1.0000	0.0000	0.0000	1.0000	1.0000

(d) Statistical Analysis for Makespan

F. Results of the 3P-GMS-ILS metaheuristic when solving the VRPD-RS-TW and related transportation systems

Tables 7 and 8 list the performance metrics of the 3P-GMS-ILS metaheuristic when solving the VRPD-RS-TW and minimizing the operational costs and makespan objective functions, respectively. Additionally, Tables 9, 10, and 11 list the performance metrics of the 3P-GMS-ILS metaheuristic when solving the VRP-TW, VRPD-TW, and VRP-RS-TW, respectively, and minimizing the operational costs objective function.

Table 7: Results of the 3P-GMS-ILS metaheuristic for the small and large ring of Amsterdam when minimizing operational costs.

Instance		Operational Costs Objective Function 3P-GMS-ILS														
#	n	ω	τ	Time [s]	T [€]	D [€]	R [€]	T [km]	D [km]	R [km]	#T	#D	#R	#ML	#RT	CH
1	5	132.23	4.24	0.32	129.91	2.32	0.00	10.86	2.32	0.00	4	1	0	1	0	140
2		57.35	1.47	0.25	45.64	4.19	7.52	6.49	4.19	7.52	0	1	4	1	0	133
3		151.39	4.94	0.31	149.91	1.48	0.00	6.58	1.48	0.00	2	3	0	1	2	153
1	10	61.41	1.45	0.95	46.50	2.67	12.24	11.78	2.67	12.24	3	1	6	1	0	162
2		133.89	4.16	1.03	127.23	4.22	2.44	9.83	4.22	2.44	6	3	1	1	2	158
3		96.30	2.33	0.96	72.62	2.89	20.79	11.12	2.89	20.79	2	3	5	1	2	160
1	15	107.13	3.18	4.22	99.03	1.66	6.44	13.98	1.66	6.44	9	1	5	1	0	178
2		168.53	5.22	4.14	165.87	0.26	2.40	36.59	0.26	2.40	13	1	1	0	1	180
3		89.38	2.26	3.44	71.80	1.07	16.51	15.81	1.07	16.51	8	2	5	1	1	155
1	50	193.21	5.31	265.00	184.47	3.12	5.62	100.10	3.12	5.62	44	2	4	2	0	468
2		201.93	5.32	259.05	187.25	12.40	2.28	110.27	12.40	2.28	46	2	2	2	0	504
3		206.97	5.58	219.24	195.08	5.19	6.70	111.20	5.19	6.70	44	3	3	3	0	489
1	75	262.19	6.32	1087.91	222.41	8.03	31.75	131.04	8.03	31.75	59	4	12	4	0	673
2		303.93	6.34	929.34	278.49	0.00	25.44	165.78	0.00	25.44	65	0	10	0	0	717
3		313.82	6.66	1120.79	307.16	0.00	6.66	202.32	0.00	6.66	71	0	4	0	0	650
1	100	854.00	9.58	100.33	854.00	0.00	0.00	616.53	0.00	0.00	100	0	0	0	0	854
2		397.67	5.64	3420.88	376.59	5.34	15.74	187.13	5.34	15.74	88	4	8	4	0	968
3		385.18	5.30	2378.72	357.02	0.00	28.16	175.47	0.00	28.16	88	0	12	0	0	777
Avg		228.69	4.74	544.27	215.05	3.05	10.59	106.83	3.05	10.59	36.22	1.72	4.56	1.28	0.44	417.66

Table 8: Results of the 3P-GMS-ILS metaheuristic for the small and large ring of Amsterdam when minimizing the makespan.

Instance		Makespan Objective Function 3P-GMS-ILS														
#	n	ω	τ	Time [s]	T [€]	D [€]	R [€]	T [km]	D [km]	R [km]	#T	#D	#R	#ML	#RT	CH
1	5	132.90	4.24	0.25	130.58	2.32	0.00	13.57	2.32	0	4	1	0	1	0	4.45
2		57.35	1.47	0.14	45.64	4.19	7.52	6.49	4.19	8	0	1	4	1	0	4.19
3		152.57	4.94	0.21	151.95	0.62	0.00	14.74	0.62	0	4	1	0	1	0	4.96
1	10	63.24	1.44	0.48	46.01	3.07	14.16	11.08	3.07	14	2	1	7	1	0	5.11
2		137.37	4.16	0.86	131.61	3.32	2.44	27.33	3.32	2	8	1	1	1	0	4.91
3		104.87	2.33	1.05	76.19	2.09	26.59	25.39	2.09	27	3	1	6	1	0	5.10
1	15	115.36	3.18	1.53	99.68	3.12	12.56	16.60	3.12	13	6	2	7	1	1	5.45
2		174.02	5.22	2.31	170.70	0.92	2.40	55.90	0.92	2	13	1	1	0	1	5.51
3		94.69	2.20	1.73	70.82	3.16	20.71	19.22	3.16	21	7	2	6	0	2	4.73
1	50	376.59	4.95	122.73	354.22	5.89	16.48	234.66	5.89	16	42	3	5	2	1	7.93
2		391.05	4.98	88.51	361.64	8.97	20.44	263.39	8.97	20	42	2	6	2	0	8.02
3		394.90	4.70	95.02	332.16	26.91	35.83	203.18	26.91	36	37	4	9	2	2	7.71
1	75	585.35	4.65	389.62	479.12	31.67	74.56	255.26	31.67	75	48	9	18	4	5	8.67
2		585.70	4.75	330.12	495.44	52.80	37.46	291.43	52.80	37	54	9	12	6	3	8.79
3		607.96	5.31	397.31	561.22	16.88	29.86	336.54	16.88	30	62	6	7	2	4	9.23
1	100	854.00	9.58	104.43	854.00	0.00	0.00	616.53	0.00	0	100	0	0	0	0	9.58
2		893.01	5.05	739.52	684.62	130.80	77.59	322.55	130.80	78	61	19	20	11	8	8.96
3		671.07	4.83	928.85	509.53	51.24	110.30	302.81	51.24	110	65	9	26	3	6	9.51
Avg		355.11	4.33	178.04	308.62	19.33	27.16	167.59	19.33	27.16	31.00	4.00	7.50	2.17	1.83	6.82

Table 9: Results of the 3P-GMS-ILS metaheuristic for the small and large ring of Amsterdam when minimizing operational costs to solve the VRP-TW.

Instance		VRP-TW					
#	n	ω	τ	Time [s]	T [€]	T [km]	#T
1	5	139.32	4.45	0.09	139.32	22.85	5
2		132.14	4.19	0.12	132.14	26.23	5
3		163.32	5.21	0.12	163.32	27.75	5
1	10	159.72	5.11	0.71	159.72	26.01	10
2		152.68	4.81	0.68	152.68	33.90	10
3		160.65	5.10	0.46	160.65	31.09	10
1	15	163.69	5.21	1.69	163.69	29.81	15
2		169.03	5.31	1.69	169.03	39.00	15
3		147.54	4.72	1.73	147.54	23.94	15
1	50	198.41	5.71	117.38	198.41	108.12	50
2		211.11	5.95	93.14	211.11	130.94	50
3		219.04	6.26	93.07	219.04	124.91	50
1	75	323.43	6.93	400.78	323.43	187.71	75
2		353.09	5.55	387.55	353.09	172.84	75
3		337.87	6.98	378.28	337.87	198.67	75
1	100	854.00	9.58	72.51	854.00	616.53	100
2		967.63	8.96	69.05	967.63	609.88	100
3		407.43	6.19	1298.10	407.43	212.38	100
Avg		292.23	5.90	162.06	292.23	145.70	42.50

Table 10: Results of the 3P-GMS-ILS metaheuristic for the small and large ring of Amsterdam when minimizing operational costs to solve the VRPD-TW.

Instance		VRPD-TW								
#	n	ω	τ	Time [s]	T [€]	D [€]	T [km]	D [km]	#T	#D
1	5	132.23	4.24	0.25	129.91	2.32	10.86	2.32	4	1
2		131.58	4.19	0.22	127.46	4.12	7.50	4.12	3	2
3		151.39	4.94	0.19	149.91	1.48	6.58	1.48	2	3
1	10	155.17	4.95	1.20	151.91	3.26	14.09	3.26	8	2
2		147.21	4.68	1.03	143.26	3.95	11.08	3.95	7	3
3		154.99	4.98	1.17	152.02	2.97	10.29	2.97	6	4
1	15	161.70	5.12	3.92	158.57	3.13	20.33	3.13	13	2
2		168.96	5.31	3.91	168.70	0.26	37.68	0.26	14	1
3		145.98	4.72	3.48	144.98	1.00	13.70	1.00	13	2
1	50	195.92	5.50	119.74	193.11	2.81	112.21	2.81	49	1
2		204.42	5.60	184.02	197.41	7.01	117.96	7.01	48	2
3		317.48	5.37	164.94	316.52	0.96	132.37	0.96	49	1
1	75	341.33	6.85	586.58	339.02	2.31	172.32	2.31	74	1
2		345.64	6.04	772.41	337.13	8.51	172.16	8.51	71	4
3		346.86	6.42	704.62	341.59	5.27	180.00	5.27	71	4
1	100	854.00	9.58	96.71	854.00	0.00	616.53	0.00	100	0
2		411.26	5.90	1284.14	396.58	14.68	187.72	14.68	94	6
3		406.10	6.46	1436.77	401.36	4.74	205.09	4.74	97	3
Avg		265.12	5.60	298.07	261.30	3.82	112.69	3.82	40.17	2.33

Table 11: Results of the 3P-GMS-ILS metaheuristic for the small and large ring of Amsterdam when minimizing operational costs to solve the VRP-RS-TW.

Instance		VRP-RS-TW								
#	n	ω	τ	Time [s]	T [€]	R [€]	T [km]	R [km]	#T	#R
1	5	139.32	4.45	0.12	139.32	0.00	22.85	0.00	5	0
2		66.07	1.75	0.19	58.55	7.52	24.39	7.52	1	4
3		163.32	5.21	0.11	163.32	0.00	27.75	0.00	5	0
1	10	69.83	1.66	0.55	55.67	14.16	23.86	14.16	3	7
2		143.62	4.44	0.71	141.18	2.44	31.90	2.44	9	1
3		101.41	2.43	0.52	80.62	20.79	30.33	20.79	5	5
1	15	118.82	3.52	1.61	112.38	6.44	27.40	6.44	10	5
2		168.60	5.22	2.00	166.20	2.40	37.91	2.40	14	1
3		91.85	2.26	1.73	74.70	17.15	27.40	17.15	9	6
1	50	198.39	5.36	104.74	187.65	10.74	107.31	10.74	44	6
2		199.41	5.31	89.82	187.35	12.06	112.09	12.06	47	3
3		205.58	5.32	97.18	193.40	12.18	135.44	12.18	42	8
1	75	270.54	6.68	401.44	238.00	32.54	150.05	32.54	63	12
2		322.78	6.19	309.33	301.83	20.95	177.26	20.95	66	9
3		323.96	6.48	429.56	312.18	11.78	165.56	11.78	69	6
1	100	854.00	9.58	69.63	854.00	0.00	616.53	0.00	100	0
2		967.63	8.96	69.20	967.63	0.00	609.88	0.00	100	0
3		385.30	5.43	807.79	368.10	17.20	186.51	17.20	93	7
Avg		266.14	5.02	132.57	255.67	10.46	139.69	10.46	38.06	4.44

G. Results of the 3P-GMS-ILS metaheuristic when solving the VRPD-RS-TW and related transportation Systems, and utilizing the same routing costs for trucks, drones, and robots

Tables 12, 13, and 14 list the performance metrics of the 3P-GMS-ILS when solving the VRPD-TW, VRP-RS-TW, and VRPD-RS-TW, respectively, and minimizing the operational costs objective function with the same routing costs for trucks, drones, and robots. Every instance was run 10 times and the best solution found is reported.

Table 12: Results of the 3P-GMS-ILS metaheuristic for the small and large ring of Amsterdam when minimizing operational costs to Solve the VRPD-TW with the same transportation costs for trucks and drones.

Instance		VRPD-TW								
#	n	ω	τ	Time [s]	T [€]	D [€]	T [km]	D [km]	#T	#D
1	5	130.41	4.24	0.44	129.74	0.68	10.19	2.70	3	2
2		128.49	4.19	0.32	127.46	1.03	7.50	4.12	3	2
3		150.28	4.94	0.28	149.91	0.37	6.58	1.48	2	3
1	10	152.52	4.95	0.95	151.39	1.13	12.01	4.52	5	5
2		143.82	4.68	1.20	142.59	1.23	8.40	4.93	5	5
3		152.32	4.98	1.15	151.17	1.15	6.87	4.61	4	6
1	15	159.29	5.11	3.67	157.78	1.52	18.47	6.06	10	5
2		167.81	5.31	1.82	166.87	0.94	30.37	3.76	9	6
3		144.09	4.68	2.05	143.06	1.02	10.69	4.09	10	5
1	50	185.89	5.23	79.18	178.11	7.79	85.22	31.14	40	10
2		193.58	5.35	164.61	185.76	7.83	100.92	31.31	41	9
3		201.61	5.49	137.00	193.40	8.22	115.33	32.86	40	10
1	75	335.84	6.72	403.91	327.63	8.21	146.22	32.85	60	15
2		337.49	6.00	482.79	331.61	5.88	154.81	23.51	63	12
3		337.63	6.14	583.35	330.24	7.39	169.71	29.57	62	13
1	100	367.16	5.76	1040.94	353.55	13.61	172.43	54.43	81	19
2		385.54	5.65	1342.63	374.43	11.10	173.63	44.41	82	18
3		392.56	5.83	1493.90	384.27	8.28	191.93	33.13	85	15
Avg		225.91	5.29	318.90	221.05	4.85	78.96	19.42	33.61	8.89

Table 13: Results of the 3P-GMS-ILS metaheuristic for the small and large ring of Amsterdam when minimizing operational costs to solve the VRP-RS-TW with the same transportation costs for trucks and robots.

Instance		VRP-RS-TW								
#	n	ω	τ	Time [s]	T [€]	R [€]	T [km]	R [km]	#T	#R
1	5	139.32	4.45	0.17	139.32	0.00	22.85	0.00	5	0
2		60.43	1.75	0.15	58.55	1.88	24.39	7.52	1	4
3		163.20	5.21	0.30	162.65	0.55	25.06	2.20	3	2
1	10	59.17	1.66	0.42	55.48	3.69	23.10	14.74	2	8
2		141.71	4.44	0.70	140.72	1.00	30.04	3.98	8	2
3		85.81	2.43	0.48	80.62	5.20	30.33	20.79	5	5
1	15	113.96	3.49	1.58	111.32	2.64	27.04	10.56	9	6
2		166.66	5.22	1.89	165.86	0.80	36.54	3.18	13	2
3		78.11	2.20	1.88	72.77	5.34	27.03	21.35	8	7
1	50	178.46	4.95	71.83	170.86	7.60	89.99	30.40	40	10
2		183.44	5.00	69.24	174.35	9.10	96.83	36.38	38	12
3		179.98	4.73	57.26	169.28	10.70	109.71	42.80	35	15
1	75	193.98	4.73	148.29	165.35	28.64	93.32	114.54	39	36
2		200.39	4.88	154.57	175.18	25.21	115.53	100.84	42	33
3		216.65	5.62	183.67	201.87	14.78	133.20	59.12	54	21
1	100	854.00	9.58	69.58	854.00	0.00	616.53	0.00	100	0
2		967.63	8.96	68.58	967.63	0.00	609.88	0.00	100	0
3		234.63	5.50	368.51	199.55	35.08	138.81	140.30	53	47
Avg		234.31	4.71	66.61	225.85	8.45	125.01	33.82	30.83	11.67

Table 14: Results of the 3P-GMS-ILS metaheuristic for the small and large ring of Amsterdam when minimizing operational costs to solve the VRPD-RS-TW with the same transportation costs for trucks, drones, and robots.

Instance		VRPD-RS-TW											
#	n	ω	τ	Time [s]	T [€]	D [€]	R [€]	T [km]	D [km]	R [km]	#T	#D	#R
1	5	130.41	4.24	0.34	129.74	0.68	0.00	10.19	2.70	0.00	3	2	0
2		48.57	1.47	0.23	45.64	1.05	1.88	6.49	4.19	7.52	0	1	4
3		150.28	4.94	0.23	149.91	0.37	0.00	6.58	1.48	0.00	2	3	0
1	10	50.23	1.45	0.74	46.50	0.67	3.06	11.78	2.67	12.24	3	1	6
2		128.75	4.16	0.94	126.89	1.06	0.81	8.46	4.22	3.22	5	3	2
3		78.54	2.33	1.00	72.62	0.72	5.20	11.12	2.89	20.79	2	3	5
1	15	100.98	3.18	2.39	98.51	0.86	1.61	11.91	3.42	6.44	7	3	5
2		165.50	5.22	1.74	163.29	1.04	1.18	26.24	4.14	4.72	6	7	2
3		74.38	2.20	2.75	68.94	1.32	4.13	11.68	5.27	16.51	5	5	5
1	50	178.39	4.95	153.72	169.22	1.67	7.51	83.42	6.66	30.02	32	6	12
2		181.86	4.98	129.88	172.37	5.16	4.34	91.88	20.63	17.34	37	5	8
3		177.11	4.70	79.28	164.27	1.14	11.70	92.55	4.56	46.79	31	2	17
1	75	192.49	4.73	289.57	162.30	7.40	22.78	81.68	29.61	91.12	34	10	31
2		198.09	4.80	415.14	170.60	5.83	21.67	106.05	23.30	86.66	40	9	26
3		217.55	5.62	466.72	197.60	2.28	17.67	116.15	9.11	70.66	45	8	22
1	100	199.40	4.59	955.72	158.29	7.27	33.85	82.54	29.07	135.40	46	10	44
2		280.60	5.67	1008.51	242.92	5.46	32.22	130.69	21.84	128.88	49	10	41
3		221.41	5.01	1044.09	178.64	4.14	38.64	113.50	16.54	154.55	45	4	51
Avg		154.14	4.12	252.94	139.90	2.67	11.57	55.72	10.68	46.27	21.78	5.11	15.61

H. Sensitivity analysis on the 3P-GMS-ILS parameters for the makespan objective function

This section presents the results of a sensitivity analysis to evaluate how variations in the 3P-GMS-ILS parameters impact the performance of the metaheuristic algorithm for the VRPD-RS-TW. The focus is on the makespan, whereas a sensitivity analysis for the operational-costs objective function is presented in Section 5.2.6. The experiments were executed using Amsterdam-based instances of 50 customers and a new set of 50 customers uniformly distributed in a square area of 20x20 [km]. Table 15 presents the average objective values and computational times of the 3P-GMS-ILS, when minimizing the makespan (τ) for combinations of $R = 25$ and $\beta = \{1.0, 2.0, 3.0\}$. Figure 1 summarizes the results of Table 15 for the objective function values and computational times. Table 16 presents the average objective values and computational times of the 3P-GMS-ILS, when minimizing the makespan for combinations of $R = \{5, 15, 25\}$ and $\beta = 3.0$. Figure 2 summarizes the results of Table 16 for the objective function values and computational times.

The results presented in Table 15 indicate that variations in the sparsification parameter have minimal impact on the performance of the 3P-GMS-ILS metaheuristic. The average objective values, denoted as τ , remain identical at 5.14 as the sparsification parameter β increases from 1.0 to 2.0 and 3.0, respectively. Furthermore, larger values of β result in a broader exploration of the solution space, leading to increased average computational times of 108.29 and 114.14 seconds for $\beta = 2.0$ and $\beta = 3.0$, respectively. From these findings, we draw the following conclusions. First, the 3P-GMS-ILS metaheuristic effectively solves the makespan optimization problem, maintaining consistently similar objective function values across different values of the sparsification parameter β . Second, the use of granular neighborhoods proved to be an efficient mechanism for avoiding unnecessary evaluations of poor-quality arcs, facilitating a more efficient exploration of the solution space.

Table 15: Analysis of 3P-GMS-ILS for the makespan objective function when varying the sparsification parameter β .

Instance			$R = 25, \beta = 1.0$		$R = 25, \beta = 2.0$		$R = 25, \beta = 3.0$	
Set	#	n	τ	Time [s]	τ	Time [s]	τ	Time [s]
Amsterdam	1	50	4.95	99.45	4.95	106.53	4.95	113.87
	2		4.98	91.43	4.98	111.51	4.98	112.87
	3		4.70	81.47	4.70	91.03	4.70	100.90
Square	1	50	5.21	89.48	5.21	108.22	5.21	114.17
	2		5.01	84.75	5.01	114.10	5.01	119.04
	3		6.02	98.11	6.01	118.33	6.01	124.02
Avg			5.14	90.78	5.14	108.29	5.14	114.14

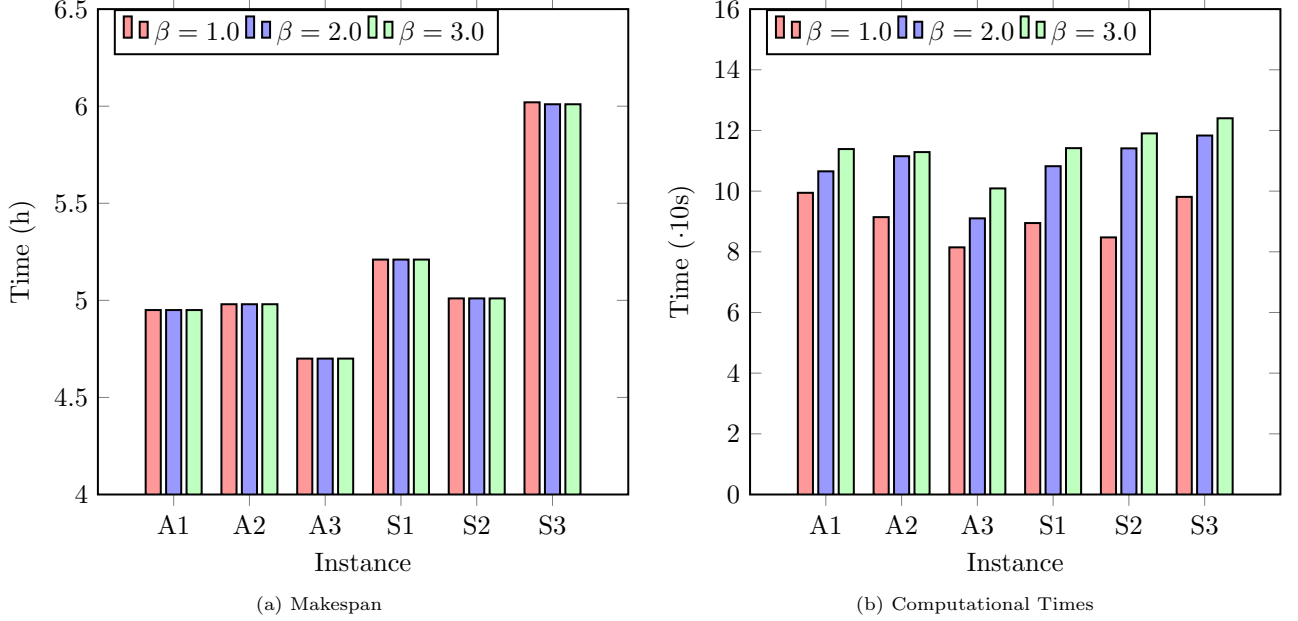


Figure 1: Comparison of makespan and computational times when $R = 15$ and $\beta = \{1.0, 2.0, 3.0\}$ for the 3P-GMS-ILS.

The results shown in Table 16 indicate that increasing the maximum number of multi-start procedures R has minimal effect on the performance of the 3P-GMS-ILS metaheuristic. Here, despite conducting a broader exploration of the solution space through additional diversification steps, the average objective values remained stable. This contrasts with the findings in Table 12, Section 5.2.6, where increasing the number of iterations led to improved average operational-costs objective values. Therefore, we conclude that the 3P-GMS-ILS effectively solves the makespan optimization problem, maintaining consistent objective values across different values of R . However, the results also reveal that higher numbers of iterations lead to significantly larger computational times, with average computational times of 36.70, 76.05, and 114.14 for R values of 5, 15, and 25, respectively. Therefore, we conclude that it is crucial to establish an appropriate balance between solution quality and computational efficiency when selecting the maximum number of iterations R in the 3P-GMS-ILS algorithm.

Table 16: Analysis of 3P-GMS-ILS for the makespan objective function when varying the number of multi-start procedures R .

Instance			$R = 5, \beta = 3.0$		$R = 15, \beta = 3.0$		$R = 25, \beta = 3.0$	
Set	#	n	τ	Time [s]	τ	Time [s]	τ	Time [s]
Amsterdam	1	50	4.95	34.24	4.95	74.50	4.95	113.87
	2		4.98	35.24	4.98	75.17	4.98	112.87
	3		4.71	29.09	4.70	68.29	4.70	100.90
Square	1	50	5.22	40.44	5.21	79.53	5.21	114.17
	2		5.03	42.60	5.01	75.49	5.01	119.04
	3		6.02	38.59	6.01	83.31	6.01	124.02
Avg			5.15	36.70	5.14	76.05	5.14	114.14

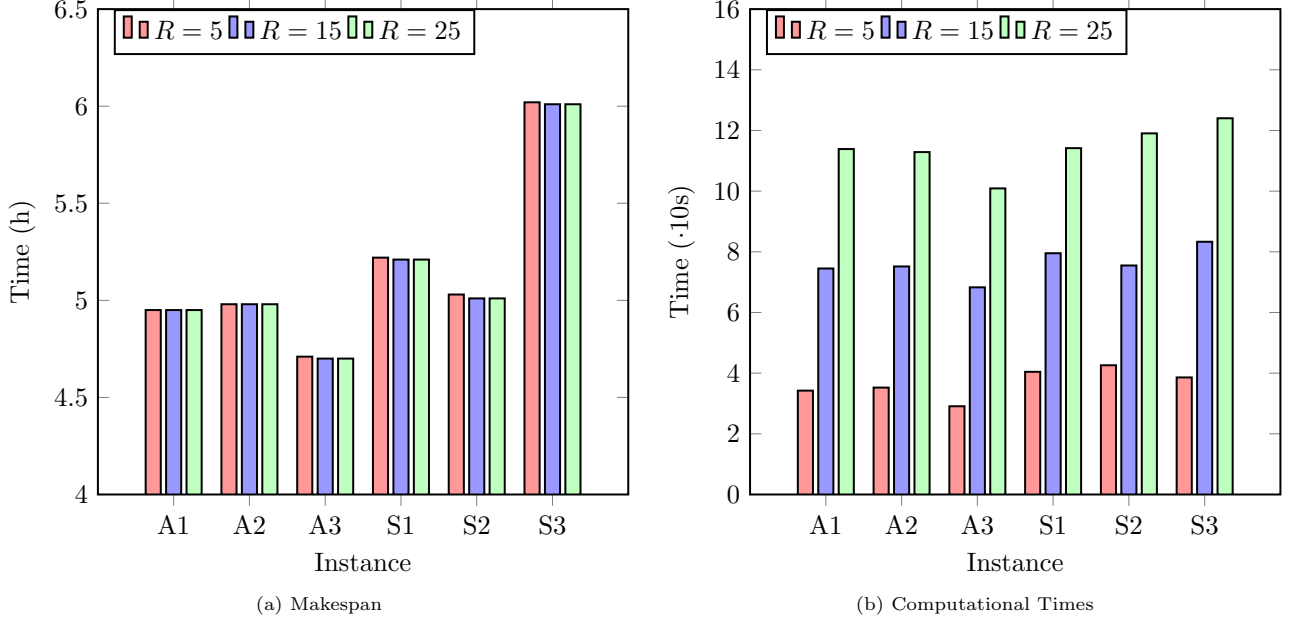


Figure 2: Comparison of makespan and computational times when $R = \{5, 15, 25\}$ and $\beta = 3.0$ for the 3P-GMS-ILS.

I. Results of the 3P-GMS-ILS for the Amsterdam set excluding energy-hovering constraints

This section presents a comparison of the 3P-GMS-ILS results on the Amsterdam set of instances when excluding the drone energy-hovering constraints. The focus is on the operational costs objective function, as the exact same results were found for the 3P-GMS-ILS when solving the makespan objective function across all the tested instances. Table 17 presents the results for the operational costs objective function (ω) when solving the set of instances, when including and excluding energy-hovering constraints. In the columns under no-hover, the table lists the percentage variation of the transportation metrics for trucks, drones, and robots, described in Table 7, computed as $\Delta = 100 \cdot \frac{old - new}{old}$. Additionally, under *no-hover*, the objective values with an asterisk represent solutions that become infeasible when the final route is evaluated considering hovering constraints.

Results show that for the small instances, i.e., 5-15 customers, the 3P-GMS-ILS metaheuristic finds the same solutions as for the case when including energy-hovering constraints. Thus, we conclude that considering energy-hovering constraints for small instances does not change the routing schedules. On the other hand, for the 50-customer instances, 3P-GMS-ILS finds different solutions when excluding energy-hovering constraints, reporting better objective function values for every tested instance. This raises the following conclusions. First, one out of the three 50-customer instances becomes infeasible when evaluating the solution under energy-hovering

constraints. This validates the importance of considering such constraints, as routing schedules excluding drone-hovering constraints might lead to battery depletion, thereby disrupting the routing schedules. Second, even though the metaheuristic used the same seeds on the same computer, the final result was different for the 50-customer instances. This demonstrates that disregarding energy-hovering constraints might not only generate new solutions, but also lead the search towards new regions of the solution space previously unexplored, as shown in instances 50-2 and 50-3.

Table 17: Analysis of 3P-GMS-ILS for the operational costs objective function excluding energy-hovering constraints.

Instance		Hovering	No-hovering									
#	n	ω	ω	ΔT [€]	ΔD [€]	ΔR [€]	ΔT [km]	ΔD [km]	ΔR [km]	$\Delta \#T$	$\Delta \#D$	$\Delta \#R$
1	5	132.23	132.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2		57.35	57.35	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
3		151.39	151.39	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1	10	61.41	61.41	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2		133.89	133.89	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
3		96.30	96.30	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1	15	107.13	107.13	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2		168.53	168.53	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
3		89.38	89.38	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
1	50	193.21	190.58*	1.03	5.13	10.14	-4.77	5.13	10.14	-2.27	-50.00	50.00
2		201.93	200.32	0.10	2.74	47.81	-0.63	2.74	47.81	0.00	-50.00	50.00
3		206.97	201.17	6.16	-225.24	81.64	-5.99	-225.24	81.64	6.82	-166.67	66.67
Avg		133.31	132.47	0.61	-18.11	11.63	-0.95	-18.11	11.63	0.38	-22.22	13.89

References

- Liu, Y. and Chen, W. (2012). A sas macro for testing differences among three or more independent groups using kruskal-wallis and nemenyi tests. *Journal of Huazhong University of Science and Technology [Medical Sciences]*, 32(1):130–134.
- Oda, T., Liu, Y., Sakamoto, S., Elmazi, D., Barolli, L., and Xhafa, F. (2015). Analysis of mesh router placement in wireless mesh networks using friedman test considering different meta-heuristics. *International Journal of Communication Networks and Distributed Systems*, 15(1):84–106.